# CONTENTS


- Abstract
- Introduction
- Problem Statement
- Methodology
- Literature Survey
- Objectiv of work
- Block Diagram
- Hardware Components
- Software Requirements
- Plan of action
- Applications
- References

# ABSTRACT


- The integration of Vision-Language Models (VLMs) in IoT systems introduces a new paradigm for smart home automation

- During the system design process, we preferred sensors that adopt low energy consumption features and allow real-time data to be transferred from residences to homeowners.
- After programming essential functions and modelling prototypes, we conducted feasibility techniques using simulation and peer-review performance tests.

- Analytics and mobile application support and make homeowners control of their residences easier.

# INTRODUCTION

- Technological advances and the acceleration of digitaltransformation have led to the emergence of needs and demandsin people's daily lives.
- These new requirements initiate the design of new technical and engineering solutions and the transformation of existing products with new concepts.
- One of the engineering solutions raised in this context is the concept of a smart home, which is formed by integrating technological features into traditional homes

# PROBLEM STATEMENT

Traditional homes lack real-time monitoring and automation, which leads to safety risks such as gas leakage, fire hazards, and thefts. Existing smart home systems in the market are often expensive and not customizable for students or middle-class families. Hence, there is a need for a low-cost IoT-based smart home system with features like:

- Safety monitoring (gas, fire, temperature sensors)
- Security features (motion detection, door/window intrusion, CCTV feed)
- Live data access via mobile app/dashboard

# METHODOLOGY

- The methodology of Smart Home Automation using IoT begins with the deployment of sensors (temperature, motion, gas, light, etc.) to collect real-time data from the home environment.
- This data is transmitted to a microcontroller or gateway device (like Arduino, Raspberry Pi, or ESP8266) for processing. Using communication protocols such as Wi-Fi, Zigbee, or Bluetooth, the processed information is sent to a cloud server for storage and analysis.
- The server enables decision-making and automation rules, which are executed by actuators or appliances in the home. Finally, users interact with the system through mobile or web applications, allowing monitoring and control of devices remotely in real time.

# LITERATURE SURVEY

| Sl. No | Paper's Referred | Author's | Year | Publication | Findings |
|---|---|---|---|---|---|
| 1 | "Development of IoT- based Smart Home Automation System withOpen-Sourced Components," | Bilsay Elis, Eylul Car, Nazli Kasap, AliThsan Sari, Atajan Rovshenov. | 2024 | ICECETEOI is 10.1109/ICECET61485.2024.10698150 | Computer and Energy Technologies (ICECET) |
| 2 | "Self-Powered IoT Device for IndoorApplications " | Ankit Kekre, Suresh K. Gawr | 2018 | ICAC3N56670.2022.10074456/10.1109 | International Conference on VLSI Design and International Conference on Embedded Systems. |
| 3 | "Home Automation System Using ESP32 andFirebase," | A. Koushal, R. Gupta, andF. Jan, | 2023 | 10.1109/ICCCT63501.2025.1 | IEEE International Conference on Smart Systems and IoT Innovations (ICSSI), |

# OBJCTIVE OF WORK

- *Remote Control & Monitoring:* Enable users to control and monitor home appliances and devices remotely using smartphones or other internet-connected devices.

- *Energy Efficiency:* Optimize energy consumption by automating lighting, heating, and cooling systems based on usage patterns and environmental conditions.

- *Enhanced Security:* Provide real-time alerts, surveillance, and access control to ensure the safety and security of the home.

- *Improved Comfort & Convenience:* Automate routine tasks, such as adjusting lighting, temperature, or scheduling appliances, to enhance the comfort and lifestyle of the users.

# BLOCK DIAGRAM

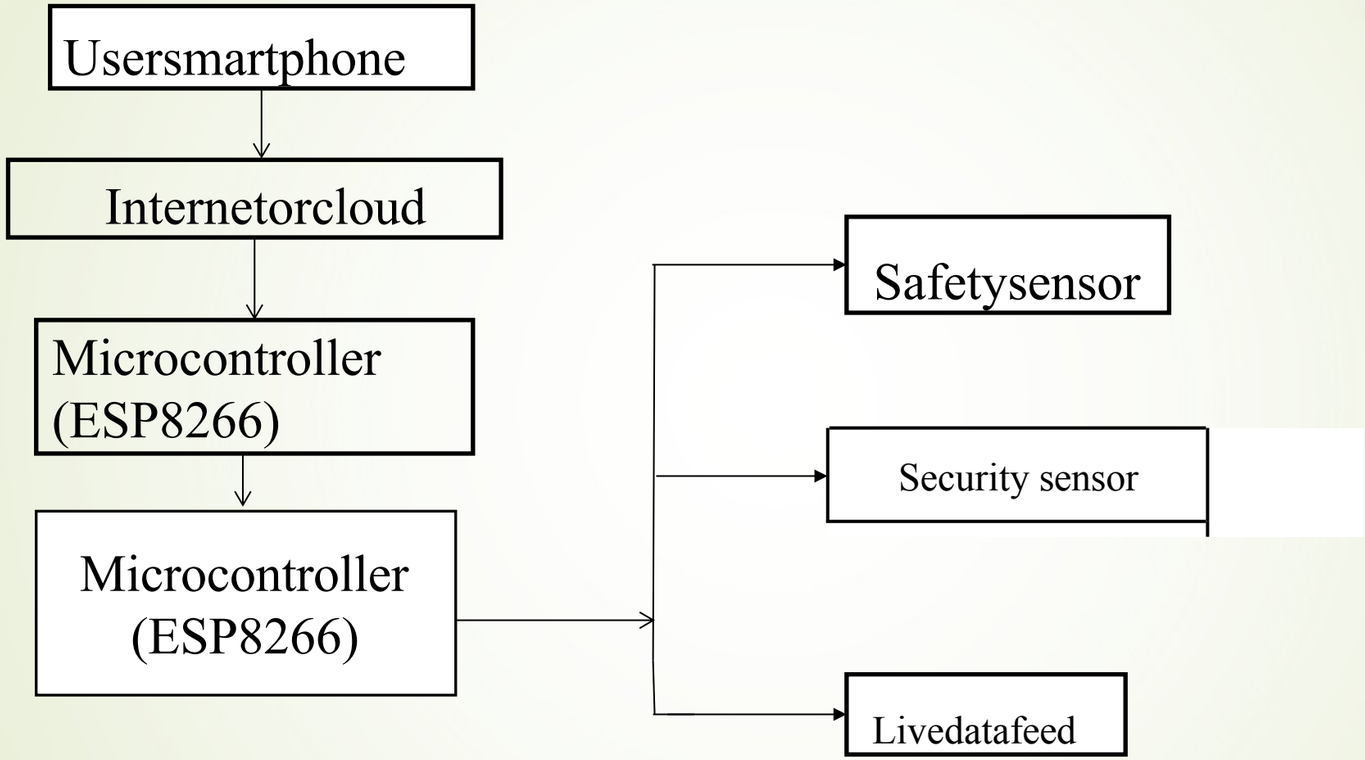
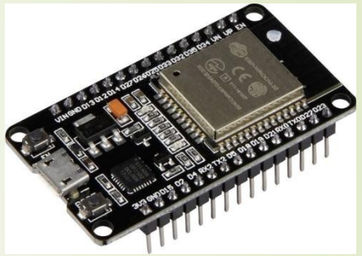



**Fig: Block diagram of Smart home automation using IOT**

# HARDWARE COMPONENTS

**1.ESP8266 / ESP32 Microcontroller:**

1. Data Processing & Control – It processes data from sensors and controls actuators like relays or alarms.
2. Wi-Fi Communication – Enables wireless communication to send sensor data to cloud servers or mobile apps.



**2.PIR Motion Sensor:**

1. *Intruder Detection* – Detects movement in a room and triggers alarms or notifications.
2. *Automatic Lighting* – Turns lights on when someone enters a space.

**3. Gas/Smoke Sensor (MQ-2):**

1. *Fire Detection* – Detects smoke or fire and triggers alerts or activates sprinklers.
2. Gas Leak Monitoring* – Identifies harmful gases like LPG and warns users to prevent accidents.

**4. Temperature & Humidity Sensor (DHT11/DHT22):**

1. *Climate Monitoring* – Tracks room temperature and humidity for comfort or safety.
2. *HVAC Control* – Adjusts fans, heaters, or coolers to maintain desired conditions.

**5. Magnetic Door/Window Sensor:**

1. *Security Monitoring* – Alerts when doors or windows are opened unexpectedly.
2. *Access Control* – Logs entries and exits or triggers alarms if unauthorized access is detected.

**6. Relay Module (for appliance control):**

1. *Switching Appliances* – Turns on/off devices like fans, lights, or pumps remotely.
2. *Automation Tasks* – Controls devices based on sensor input (e.g., turn off a heater if temperature is high).

**7.Camera Module (ESP32-CAM or IP Camera:**

1. *Surveillance* – Provides live video feeds for monitoring security
2. *Motion Detection Recording* – Captures footage when movement is detected.

**8. Buzzer & LEDs :**

1. *Alerts and Alarms* – Sounds alarms and flashes LEDs in emergency situations.
2. *Status Indicators* – Shows system health or operation mode.

# SOFTWAREREQUIREMENTS

- ArduinoIDE/PlatformIO
- MobileApp(Blynk)
- WebDashboardformonitoring
- CloudBasedHubs

# PLAN OF ACTION

| SL NO | MONTH | ACTIVITY |
|---|---|---|
| 1 | August/2025 | Collecting journals and literature survey,study on hardware components |
| 2 | september/2025 | Synopsis and purchasing components |
| 3 | October/2025 | Implementation and testing |
| 4 | November/2025 | Making of report |
| 5 | November/2025 | Demonstration of project |

# APPLICATIONS

- Residential Home
- Office Buildings
- Hospitals (for patient safety monitoring)
- Hotels & Apartments
- Smart Cities

# REFERENCES


- [1] Bilsay Elis, Eylul Car, Nazli Kasap, Ali Thsan Sari, Atajan Rovshenov, "Development of IoT-based Smart Home Automation System with Open-Sourced Components," 2024 7th International Conference on Electrical, Computer and Energy Technologies (ICECET), 2024, pp. 254-260, doi: 10.1109/ICECET.2024.12345110.
- Ankit Kekre, Suresh K. Gawre," Rolf Arne Kjellby, Thor Eirik Johnsrud, Svein Erik Loetveit,Linga Reddy Cenkeramaddi,Mohamed Hamid and Baltasar Beferull-Lozano,"Self-Powered IoT Device for Indoor Applications", 2018 31thInternational Conference on VLSI Design and 2018 17thInternational Conference on Embedded Systems.
- [3] A. Koushal, R. Gupta, and F. Jan, "Home Automation System Using ESP32 and Firebase," in Proceedings of the IEEE International Conference on Smart Systems and IoT Innovations (ICSSI), 2023.

# Thank You